Project Design Phase
Proposed Solution Template

Date	12-10-2026
Team ID	SWUID2026402745
Project Name	HotelPro - Hotel Room Booking and Management System
Maximum Marks	2 Marks

Proposed Solution for HotelPro

S. No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Customers struggle to find and filter available hotel rooms in one place, verify real-time availability and get instant confirmations, while manual front-desk registers and scattered booking channels cause long check-in/check-out queues, double bookings, unclear cancellations and delayed refunds. Hotel staff and administrators lack a unified platform to manage room inventory, guest records, billing and live occupancy, leading to operational inefficiencies and lost revenue.
2	**Idea / Solution Description**	HotelPro is a full-stack web application (React.js and Bootstrap frontend + Node.js/Express.js REST API backend + MongoDB via Mongoose) where customers search rooms by city, dates and guest count, filter by price, room type, amenities and rating, book online with automated price and tax calculation, and pay through an integrated payment gateway (card, UPI, net banking). Room availability is re-verified at confirmation and enforced with compound database indexing so two users never double-book the same room. Hotel staff and administrators manage hotels, rooms, guest ID proofs, check-in/check-out workflows, refunds and revenue analytics through JWT-secured role-based portals.
3	**Novelty / Uniqueness**	- Real-time room availability engine with confirmation-time re-verification and compound indexing to eliminate double bookings - Unified three-tier platform connecting customer self-service booking, front-desk reception operations and property administration - Policy-based online booking modification and cancellation before check-in with automated room release and refund tracking

S. No.	Parameter	Description
		- Digital front-desk check-in and check-out console that logs actual timestamps, verifies balances and auto-updates room states (Available, Occupied, Maintenance) - Complete guest lifecycle management linking primary and accompanying guest details, ID proofs and stay histories to reservations - Live administrator dashboard with occupancy rates, booking trends, revenue analytics and downloadable CSV/PDF reports
4	**Social Impact / Customer Satisfaction**	- Saves time for travelers and hotel receptionists by eliminating manual paperwork and front-desk queues from any device - Transparent nightly pricing, tax calculations, payment receipts and clear booking/refund statuses build customer trust - Empowers independent hotels and property owners with an affordable, modern digital reservation and management system - Protects guest personal data and payment security using bcrypt password hashing, HTTPS, JWT RBAC and zero raw card storage - Prevents reservation conflicts and arrival surprises through instant booking confirmations and accurate inventory control
5	**Business Model (Revenue Model)**	- Platform convenience or service fee on every online room reservation completed through the portal - Commission from partnered hotels on confirmed bookings and online payments processed via the platform - Featured hotel and room listings on the home/landing page for higher visibility in city search results - Additional service charges for early check-in, late check-out and premium room upgrades (Deluxe, Suite) - SaaS subscription tiers for hotel administrators offering multi-property management and advanced CSV/PDF BI analytics
6	**Scalability of the Solution**	- Stateless JWT-secured Node.js/Express.js REST API that scales horizontally behind a load balancer to handle 500+ concurrent users - MongoDB Atlas cloud deployment supporting replica sets, sharded clusters and compound indexes (email, hotelId, roomId, dates) for sub-2-second queries - Modular MVC-style codebase (routes, controllers, Mongoose models, middleware) for seamless integration of payment gateways and email/SMS alerts

S. No.	Parameter	Description
		- Extensible architecture ready for future native Android/iOS apps, OTA channel managers (Booking.com, Expedia), dynamic pricing and multi-currency support